

Perturbation Catalogue

An Integrated Resource for Exploring the Impact of Genetic Perturbations

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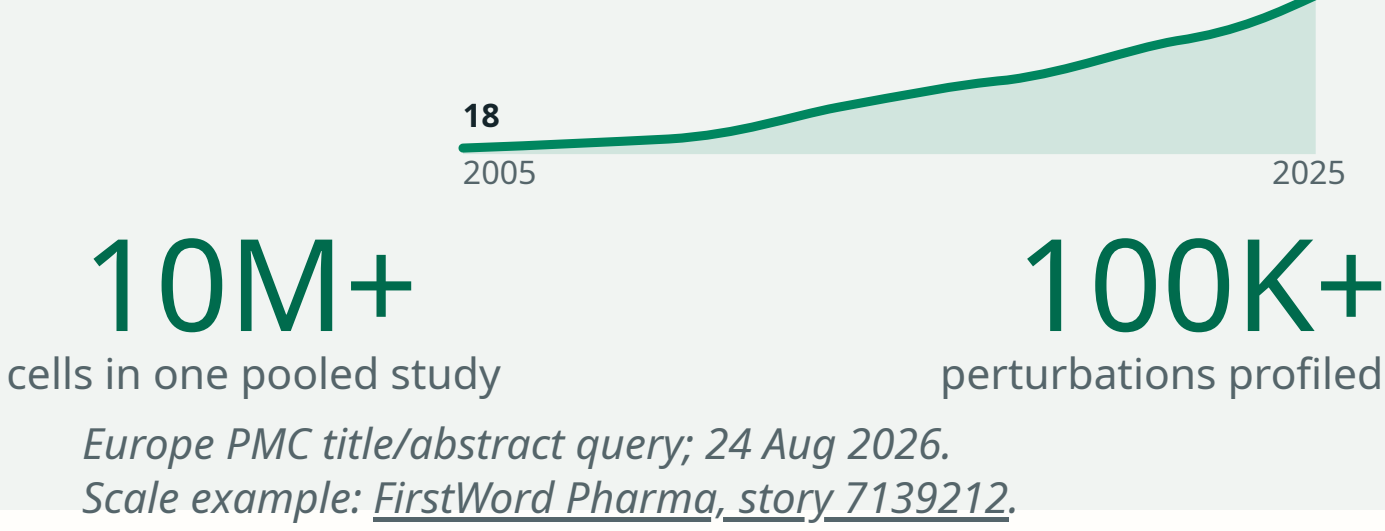
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Perturbation evidence is multiplying faster than it can be organised

A single pooled genome-scale Perturb-Seq dataset can **simultaneously profile tens of millions of cells across hundreds of thousands of perturbations**. That scale could reveal robust biology across studies, but identifiers, metadata, assays and processing pipelines differ. Results remain hard to compare or reproduce. The Catalogue aims to supply a shared schema, stable gene identity and one access path for combining this evidence.

36× growth
in perturbation papers
over two decades



Our approach turns scattered datasets into a unified resource

1,237
datasets

19,235
targets

36
tissues

34
cell types

1,200
cell lines

169
diseases

CRISPR screens · 1,201 datasets

Pooled guides → gene-level phenotypes and fitness.

Perturb-Seq · 26 datasets

Guide identity + scRNA-seq → cell-state responses.

MAVE · 10 datasets

Variant libraries + assays → sequence-function maps.

We extensively curate metadata...

84
meta-
data
fields

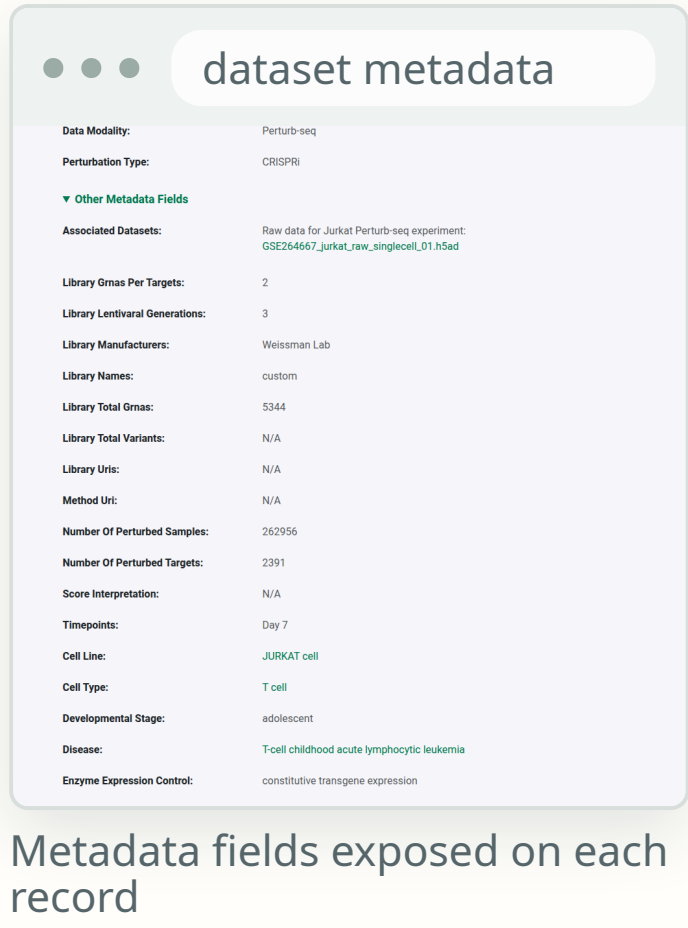
- Study & provenance**

Title, authors, year, URI, licence
- Model system**

Organism, tissue, cell line, disease
- Perturbation & library**

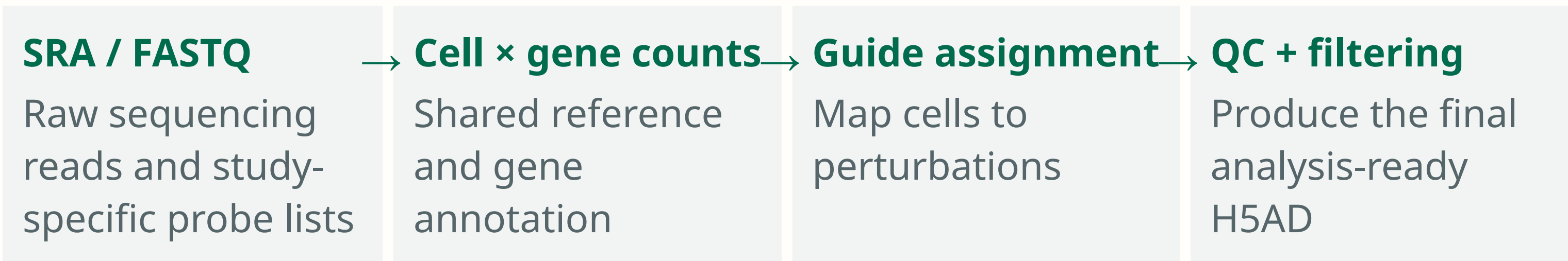
Targets, guides, library, variants
- Readout & processing**

Assay, technology, kit, reference



And reprocess data from scratch

Author-provided data differ in format, alignment, annotation, QC, filtering and guide assignment. Selected Perturb-Seq studies follow one shared computational path.



Stable Ensembl gene IDs + ontology terms make cross-study questions searchable.

Browse the Catalogue

Search → Context → Filter → Explore

Targets

Search by BRCA2

Dataset: Nadig 2025 Jurkat

Access records via API

Structured JSON for custom pipelines, reproducible analysis and machine learning.

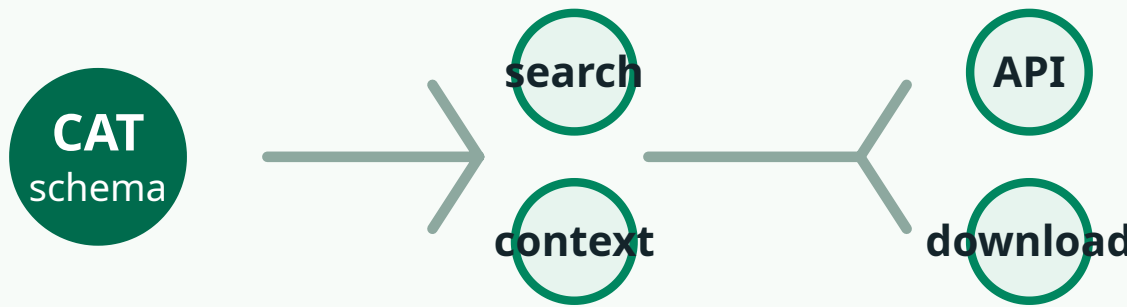
```
GET /v1/perturb-seq/nadig_2025_jurkat/search?perturbation_gene_name=TFAM&limit=1

{
  "perturbation": {
    "gene_symbol": "TFAM",
    "effect": {
      "gene_symbol": "MT-ND5",
      "log2fc": -1.926,
      "padj": 0.0
    }
  }
}
```

Download entire datasets

Data is available in **Parquet** and **CSV.gz**, and metadata as **JSON**.

One Catalogue, three access routes



Stay tuned for more exciting things coming soon

- Extract structured metadata faster**

AI-assisted drafting with curator review and validated ontology mapping.
- Reanalyse more designs**

More kits, experimental structures and independent quality metrics.
- Publish regular, versioned releases**

Explicit provenance and stable snapshots for reproducible downstream use.
- Reanalyse MAVE data**

Enrich variant-to-function coverage with harmonised scores.

Explore: ebi.ac.uk/perturbation-catalogue Get in touch: perturbation-catalogue-help@ebi.ac.uk